

Gridless Synthetic Aperture Sonar Imaging Based on Atomic Norm Minimization

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Abstract—Synthetic Aperture Sonar (SAS) is a key technology for high-resolution underwater imaging. However, traditional imaging algorithms, such as the Range-Doppler (RD) algorithm and standard Compressive Sensing (CS) methods, typically rely on discretized grids. This dependence inevitably leads to basis mismatch and grid quantization errors when targets do not align perfectly with the predefined grid points, thereby degrading imaging precision and resolution. To address this limitation, a novel gridless imaging method based on Atomic Norm Minimization (ANM) is proposed in this paper. By formulating the azimuth reconstruction as a continuous frequency recovery problem, the proposed framework effectively eliminates grid-induced errors. Numerical simulations demonstrate that the proposed method significantly outperforms traditional grid-based algorithms, achieving superior super-resolution capability and reconstruction accuracy in sparse sampling scenarios.

Index Terms—Synthetic Aperture Sonar (SAS), Atomic Norm Minimization (ANM), gridless imaging, sparse sampling.

I. INTRODUCTION

Synthetic Aperture Sonar (SAS) has emerged as a powerful technique for high-resolution underwater imaging, capable of providing constant along-track resolution independent of range and frequency [1]–[3]. Due to its ability to generate high-quality imagery of the seabed, SAS is extensively applied in diverse fields, including mine countermeasures, pipeline inspection, and underwater archaeology [4]–[6]. The increasing demand for precise environmental monitoring and target identification has fueled significant research enthusiasm in developing advanced SAS signal processing algorithms, aiming to enhance imaging fidelity and operational efficiency in complex underwater environments [7], [8].

Traditionally, SAS imaging relies on matched-filtering based algorithms, such as the Range-Doppler (RD) algorithm and the Back-Projection (BP) algorithm [9], [10]. While computationally efficient, these methods are constrained by the Nyquist sampling theorem, requiring dense spatial sampling to avoid aliasing artifacts. To address data volume and acquisition time constraints [11], Compressive Sensing (CS) techniques based on ℓ_1 -norm minimization have been introduced. These methods exploit the spatial sparsity of underwater targets to reconstruct images from undersampled data. In [11], a CS-based approach was introduced to achieve high-resolution SAS

imaging with reduced sampling requirements, proving robust even in scenarios with sensor loss. In [12], SAS imaging is formulated as an ℓ_1 -regularized least-squares problem, efficiently solved via a distributed ADMM algorithm to enhance resolution. However, standard CS approaches typically assume that targets lie exactly on a predefined discrete grid.

The reliance on a fixed grid in conventional CS-based methods introduces a fundamental limitation known as "basis mismatch" or "grid quantization error" [13]. In practical scenarios, target positions are continuous and rarely align perfectly with the discretized grid points. This mismatch leads to leakage of signal energy into adjacent grid cells, resulting in inaccurate intensity estimation and reduced resolution. Although refining the grid can mitigate this error, it inevitably increases the coherence of the sensing matrix and the computational burden, often leading to numerical instability. Therefore, there is a critical need for a reconstruction framework that can operate directly in the continuous domain to overcome these grid-induced limitations [14], [15].

To address these challenges, this paper proposes a novel gridless imaging method for SAS based on Atomic Norm Minimization (ANM). Specifically, the proposed approach leverages a dechirp operation to transform SAS echo signals into a representation suitable for ANM-based optimization. By formulating the azimuth reconstruction as a continuous frequency recovery problem using the Vandermonde decomposition of Toeplitz matrices, the proposed method effectively circumvents the basis mismatch and grid quantization errors inherent in discretized approaches. Unlike the traditional RD algorithm bound by the Rayleigh limit, and conventional grid-based CS methods suffering from basis mismatch, our ANM-based approach demonstrates superior super-resolution capabilities. Simulation results further validate that the proposed method maintains high reconstruction fidelity even under severe undersampling conditions, offering a robust solution for high-precision sparse SAS imaging.

II. SIGNAL MODEL

The geometry of SAS is depicted in Fig. 1. The direction of y along which the sonar moves is defined as azimuth or the cross-range axis, and the direction x perpendicular to y is the slant-range axis. Assuming the platform moves along the cross-range axis, it synthesizes scattered signals from the target

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by transmitting and receiving signals, thereby generating an underwater image. The transmitted linear frequency modulated (LFM) signal is given by

$$s_t(\hat{t}) = \text{rect}\left(\frac{\hat{t}}{T_p}\right) \exp(j2\pi f_c \hat{t} + \pi \gamma \hat{t}^2), \quad (1)$$

where $\hat{t}$ denotes fast time. T_p , f_c , and γ represent the pulse duration, carrier frequency, and chirp rate, respectively. Also, the window function is

$$\text{rect}(x) = \begin{cases} 1, & |x| \leq \frac{1}{2} \\ 0, & |x| > \frac{1}{2}. \end{cases} \quad (2)$$

Consider a scenario with P point targets. The coordinates of the p th target are (x_p, y_p) , and its echo delay is

$$\tau_p(t_m) = \frac{2R_p(t_m)}{c}, \quad (3)$$

where $R_p(t_m)$ stands for the instantaneous slant range between the p th target and the receiving hydrophone at time t_m , and c is the speed of sound. Consider a raw echo dataset consisting of N range samples and M azimuth pulses. After demodulation to baseband, the echo signal can be expressed as

$$s_r(\hat{t}, t_m) = \sum_{p=1}^P \delta_p \text{rect}\left(\frac{\hat{t} - \tau_p(t_m)}{T_p}\right) \omega_a(t_m - \frac{y_p}{v_r}) \times \exp(-j2\pi f_c \tau_p(t_m)) \times \exp(j\pi \gamma (\hat{t} - \tau_p(t_m))^2), \quad (4)$$

where t_m denotes slow time, δ_p is the backscatter coefficient of the p th target, ω_a is the azimuth envelope, y_p is the azimuth position of the p th target, and v_r is the velocity of the platform.

After range compression and range cell migration correction (RCMC), the echo signal can be expressed

$$\begin{aligned} s_{rc}(\hat{t}, t_m) &= \sum_{p=1}^P \delta_p p_r(\hat{t} - \frac{2R_{0,p}}{c}) \omega_a(t_m - t_{c,p}) \\ &\quad \exp(-j\frac{4\pi f_c R_{0,p}}{c}) \exp(-j4\pi f_c \frac{v_r^2(t_m - t_{c,p})^2}{2R_{0,p}c}) \\ &= \sum_{p=1}^P \delta_p p_r(\hat{t} - \frac{2R_{0,p}}{c}) \omega_a(t_m - t_{c,p}) \\ &\quad \times \exp(-j\frac{4\pi f_c R_{0,p}}{c}) \exp(-j\frac{2\pi f_c v_r^2}{R_{0,p}c} t_m^2) \\ &\quad \times \exp(j\frac{4\pi f_c v_r^2 t_{c,p}}{R_{0,p}c} t_m) \exp(-j\frac{2\pi f_c v_r^2 t_{c,p}^2}{R_{0,p}c}), \end{aligned} \quad (5)$$

where

$$t_{c,p} = \frac{y_p}{v_r} \quad (6)$$

and $R_{0,p}$ refers to the slant range when the SAS boresight is directed to the p th target.

Then, we perform azimuth dechirp, with the reference function set to

$$s_{ref}(\hat{t}, t_m) = \exp(j\frac{2\pi f_c v_r^2}{R_{0,p}c} t_m^2). \quad (7)$$

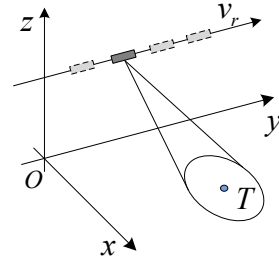


Fig. 1: Geometric model of the monostatic synthetic aperture sonar system.

Multiplying this by the signal yields

$$\begin{aligned} y(\hat{t}, t_m) &= s_{rc}(\hat{t}, t_m) s_{ref}(\hat{t}, t_m) \\ &= \sum_{p=1}^P \delta'_p \exp(j2\pi (\frac{2v_r^2 t_{c,p}}{\lambda R_{0,p}}) t_m) \\ &= \sum_{p=1}^P \delta'_p \exp(j2\pi (\frac{2v_r^2 t_{c,p}}{\lambda R_{0,p}}) \frac{m}{\text{PRF}}) \\ &= \sum_{p=1}^P \delta'_p \exp(j2\pi (\frac{2v_r y_p}{\lambda R_{0,p} \text{PRF}}) m), \end{aligned} \quad (8)$$

where

$$\begin{aligned} \delta'_p &= \delta_p p_r(\hat{t} - \frac{2R_{0,p}}{c}) \omega_a(t_m - t_{c,p}) \\ &\quad \times \exp(-j(\frac{4\pi f_c R_{0,p}}{c} + \frac{2\pi f_c v_r^2 t_{c,p}^2}{R_{0,p}c})). \end{aligned} \quad (9)$$

PRF is the pulse repetition frequency.

III. PROPOSED IMAGING METHOD

A. Atomic Norm Minimization Formulation

Considering the n th range cell, let $\mathbf{y} \in \mathbb{C}^M$ denotes the azimuth measurement vector consisting of M observations. The discretized signal is given by

$$\mathbf{y}_m = \sum_{p=1}^P c_p e^{j2\pi f_p m} + w_m, \quad m \in \{0, \dots, M-1\}, \quad (10)$$

where $c_p = \delta'_p$, w_m represents the additive noise and

$$f_p = \frac{2v_r y_p}{\lambda R_{0,p} \text{PRF}}. \quad (11)$$

The steering vector $\mathbf{a}(f_p) = [1, e^{j2\pi f_p}, \dots, e^{j2\pi(M-1)f_p}]^T \in \mathbb{C}^M$ is defined as an atom of the signal, and the steering/mapping matrix is $\mathbf{A} = [\mathbf{a}(f_1), \dots, \mathbf{a}(f_p)] \in \mathbb{C}^M$. The notation $\mathbb{C}$ denotes the set of complex numbers. Then, the set of atoms $\mathcal{A}$ in this application is defined as

$$\mathcal{A} = \{\mathbf{a}(f_p) : f \in [0, 1]\}. \quad (12)$$

The atomic norm $\|\mathbf{y}\|_{\mathcal{A}}$ associated with the atomic set $\mathcal{A}$ is given by

$$\|\mathbf{y}\|_{\mathcal{A}} = \inf \left\{ \sum_{p=1}^P c_p : \mathbf{y} = \sum_{p=1}^P c_p \cdot \mathbf{a}(f_p), \mathbf{a}(f_p) \in \mathcal{A} \right\}. \quad (13)$$

Gridless Compressive Sensing aims to recover the continuous-valued frequencies from measurements $\mathbf{y}$ corrupted by additive noise. This problem can be formulated as an ANM problem [16]

$$\min_{\tilde{\mathbf{y}}} \mu \cdot \|\tilde{\mathbf{y}}\|_{\mathcal{A}} + \frac{1}{2} \|\mathbf{y} - \tilde{\mathbf{y}}\|_2^2. \quad (14)$$

The atomic norm minimization problem in (14) can be equivalently recast as the following semidefinite program (SDP)

$$\begin{aligned} \min_{\tau, \mathbf{u}, \tilde{\mathbf{y}}} \quad & \mu \left(\frac{\tau}{2} + \frac{1}{2M} \cdot \text{tr}(T(\mathbf{u})) \right) + \frac{1}{2} \|\mathbf{y} - \tilde{\mathbf{y}}\|_2^2 \\ \text{subject to} \quad & \begin{bmatrix} \tau & \tilde{\mathbf{y}}^H \\ \tilde{\mathbf{y}} & T(\mathbf{u}) \end{bmatrix} \geq 0 \end{aligned} \quad (15)$$

where $\mathbf{u} \in \mathbb{C}^M$ and $T(\mathbf{u}) \in \mathbb{C}^{M \times M}$ denotes a Toeplitz matrix with

$$T(\mathbf{u}) = \begin{bmatrix} u_1 & u_2 & \dots & u_M \\ u_2^H & u_1 & \dots & u_{M-1} \\ \vdots & \vdots & \ddots & \vdots \\ u_M^H & u_{M-1}^H & \dots & u_1 \end{bmatrix}, \quad (16)$$

where u_i denotes the i th entry of $\mathbf{u}$. The atomic-norm optimization problem (15) can be computed by the SDP algorithm. The optimization problem can be efficiently solved using CVX toolbox with solvers such as SDPT3.

B. Azimuth and Reflectivity Recovery

After solving the SDP problem in (15), we obtain the optimal Toeplitz matrix $T(\mathbf{u})$ and the denoised signal $\tilde{\mathbf{y}}$. The regularization parameter is set as $\mu \propto \sigma \sqrt{M \log M}$ with noise standard deviation σ , following [16]. According to the Vandermonde decomposition theorem, $T(\mathbf{u})$ can be uniquely decomposed as

$$T(\mathbf{u}) = \sum_{p=1}^P |c_p|^2 \mathbf{a}(f_p) \mathbf{a}(f_p)^H. \quad (17)$$

Here, the number of targets $\hat{P}$ is estimated by counting the dominant eigenvalues of $T(\mathbf{u})$ that exceed a predefined noise threshold. Subsequently, the Root-MUSIC algorithm is employed to extract the frequency estimates $\{\hat{f}_p\}_{p=1}^{\hat{P}}$ from $T(\mathbf{u})$. The azimuth positions $\hat{y}_p$ are retrieved via

$$\hat{y}_p = \frac{\hat{f}_p \lambda R_{0,p} \text{PRF}}{2v_r}. \quad (18)$$

With the estimated frequencies $\{\hat{f}_p\}_{p=1}^{\hat{P}}$, the steering matrix $\hat{\mathbf{A}}$ can be written as

$$\hat{\mathbf{A}} = [\mathbf{a}(\hat{f}_1), \mathbf{a}(\hat{f}_2), \dots, \mathbf{a}(\hat{f}_{\hat{P}})] \in \mathbb{C}^{M \times \hat{P}}. \quad (19)$$

Finally, the complex reflectivities $\hat{\mathbf{c}}$ are obtained by solving the overdetermined system $\hat{\mathbf{y}} = \hat{\mathbf{A}} \hat{\mathbf{c}}$ via

$$\hat{\mathbf{c}} = (\hat{\mathbf{A}}^H \hat{\mathbf{A}})^{-1} \hat{\mathbf{A}}^H \hat{\mathbf{y}}. \quad (20)$$

IV. SIMULATION

In this section, numerical simulations are conducted to validate the effectiveness of the proposed ANM-based imaging method. We assess the performance in terms of sparse recovery accuracy under undersampled conditions and super-resolution

capability, comparing it against the traditional RD algorithm and the grid-based CS method [12].

A monostatic SAS system operating at a carrier frequency of $f_c = 100$ kHz with a bandwidth of $B = 10$ kHz is considered in the simulation environment. The detailed system parameters are listed in Table I. The echo data is corrupted by additive white Gaussian noise to simulate a realistic underwater acoustic environment. To evaluate the reconstruction fidelity, a scene containing nine ideal point targets is constructed. These targets are arranged in a 3×3 grid pattern with identical reflectivity coefficients, located at coordinates spanning from 49.5 m to 50.5 m in the slant range and from -0.5 m to 0.5 m in the cross range.

TABLE I: Parameters Used for Simulation Results

Parameters	Values
Frequency, f_c	100 kHz
Pulse duration, T_p	10 ms
Bandwidth, B	10 kHz
Platform velocity, v_r	3 m/s
Transducer size (monostatic)	0.2 m
Pulse repetition frequency, PRF	120 Hz

We first assess the robustness of the proposed method against the RD algorithm and the grid-based CS method under 50% random undersampling conditions. Figure 2 (a) illustrates the ground truth of the point targets, serving as the reference for reconstruction evaluation. The imaging result of the RD algorithm is shown in Fig. 2 (b). Due to the violation of the Nyquist sampling theorem, the RD algorithm suffers from significant degradation, exhibiting severe aliasing artifacts and elevated sidelobes that blur the targets. Figure 2 (c) presents the reconstruction result of the grid-based CS method. By exploiting the sparsity prior via l_1 -norm minimization, the CS method effectively suppresses the aliasing artifacts observed in the RD result. However, due to the inherent basis mismatch, the target energy is not perfectly concentrated, leading to slight energy leakage into adjacent grid cells. In contrast, Fig. 2 (d) demonstrates that the proposed ANM-based method achieves the best focusing performance. By recovering frequencies in the continuous domain, the proposed method not only eliminates grid-induced energy leakage but also preserves high fidelity with highly focused target energy that closely matches the ground truth in Fig. 2 (a).

To quantitatively evaluate the reconstruction accuracy, Fig. 3 illustrates the root mean square error (RMSE) of the azimuth position estimates versus the signal-to-noise ratio (SNR). The RD algorithm shows limited accuracy due to the Rayleigh resolution limit. The grid-based CS method significantly improves accuracy compared to RD algorithm. However, a critical observation is that its RMSE curve plateaus at high SNR levels. This phenomenon indicates a performance floor caused by irreducible grid quantization errors. In contrast, the proposed ANM method consistently achieves the lowest RMSE. By performing recovery in the continuous frequency domain, the proposed method circumvents the off-grid limitations. Consequently, its RMSE continues to decrease as SNR improves, demonstrating its superior capability in achieving

high-precision reconstruction without the bottleneck of basis mismatch.

Furthermore, to further investigate the gridless feature and resolution limit of the proposed algorithm, a specific experiment is designed with two closely spaced targets separated by only 0.14 m in the azimuth direction. Figure 4 displays the imaging results for the range cell where the targets are located. As observed, the RD algorithm fails to distinguish the targets, conflating them into a single lobe due to the Rayleigh limit. The grid-based CS method resolves two peaks but exhibits position deviations caused by basis mismatch. In contrast, the proposed ANM method perfectly distinguishes the targets with estimates aligning precisely with the ground truth, validating its superior super-resolution capability.

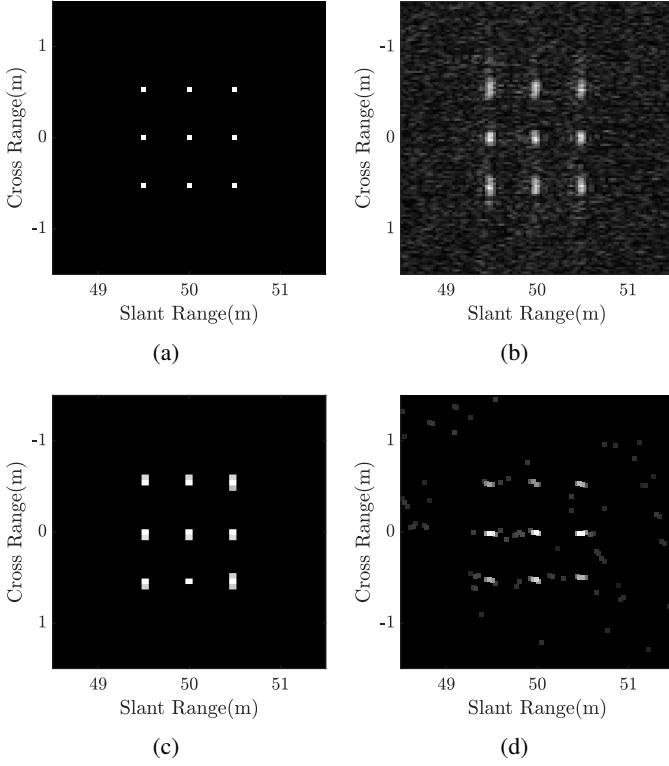


Fig. 2: Comparison of imaging results at -10 dB SNR. (a) Ground truth of the point targets. (b) RD algorithm with 50% random undersampling. (c) CS method with 50% random sampling. (d) Proposed ANM method with 50% random undersampling.

Apart from the imaging performance, we briefly discuss the computational complexity and scaling of the evaluated algorithms. The grid-based ADMM-CS method entails an approximate computational complexity of $\mathcal{O}(KMN_g)$, where K , M , and N_g represent the number of iterations, azimuth observations, and discrete grid pixels, respectively. In contrast, the proposed gridless ANM method is formulated as an SDP problem and solved via interior-point methods (e.g., CVX/SDPT3), which yields a theoretical complexity of $\mathcal{O}(M^{4.5})$ per range cell. This indicates that the computational burden of the proposed method scales polynomially with

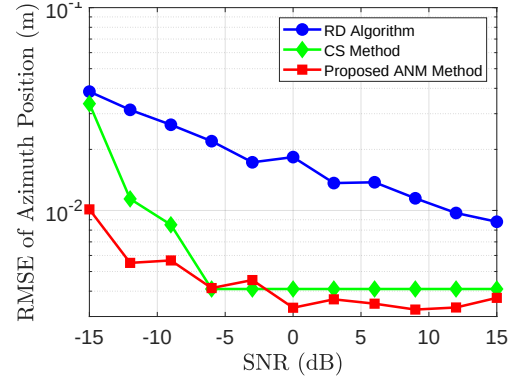


Fig. 3: RMSE of azimuth position versus SNR for the RD algorithm, CS method and the proposed method.

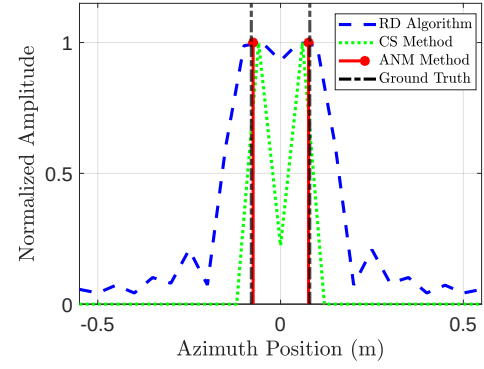


Fig. 4: Azimuth profiles of two targets spaced 0.14 m apart using three algorithms.

the synthetic aperture length. However, because the ANM framework is highly robust to sparse sampling scenarios, it can operate effectively with a significantly reduced number of observations. This capability to maintain high precision under severe undersampling effectively mitigates the computational scaling associated with long apertures, rendering the proposed method computationally feasible in practice.

V. CONCLUSION

This paper presents a novel gridless imaging method for SAS based on ANM. By formulating the azimuth reconstruction as a continuous frequency recovery problem, the proposed approach effectively circumvents the basis mismatch and grid quantization errors inherent in traditional discrete methods. Numerical simulations demonstrate the superiority of the proposed method over the conventional RD algorithm and CS method in robustness, reconstruction accuracy, and super-resolution capability. These results suggest that the ANM-based framework is a promising candidate for high-precision underwater imaging, particularly in scenarios requiring sparse sampling and super-resolution target distinction. In practice, motion errors may introduce phase perturbations that degrade the Toeplitz structure of the signal. We will investigate these practical effects and their compensation methods in future work.

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